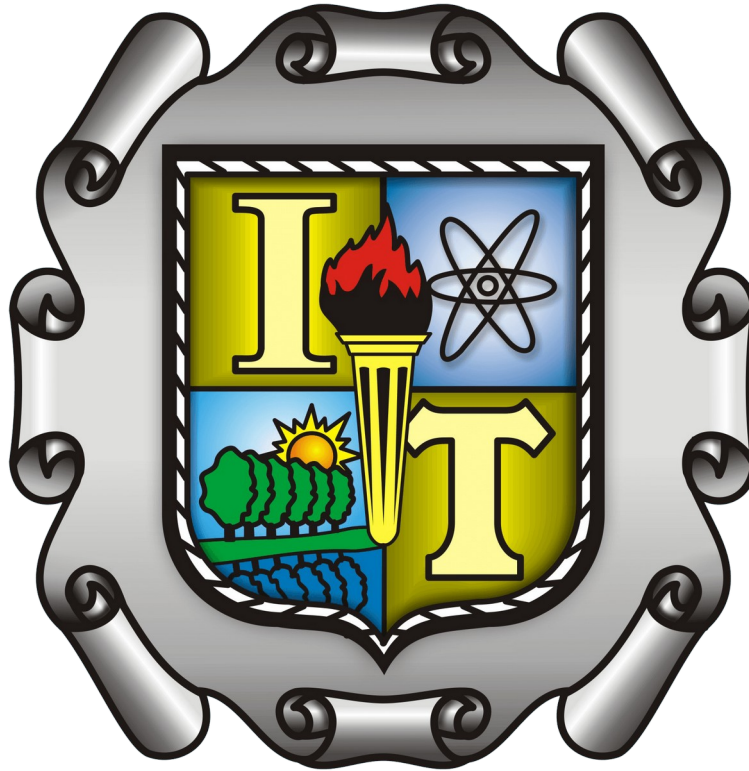


Tecnologico nacional de mexico

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computo en la nube

5.4 Instalación de Docker

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Instalar docker en debiant, aquí hubo problemas con el ssh pero se resolvio porque faltaba un permiso por parte de la maquina host

```
Archivo  Editar  Vista  felipe@pop-os: ~

gstreamer1.0-plugins-ugly  libsqlite3-0          sudo
gstreamer1.0-x            libss2               tzdata
ifupdown                 libssl3t64          wireless-regdb
imagemagick-7-common     libtalloc2
inetutils-telnet         libtdb1

Installing dependencies:
  libaio1t64  linux-image-6.12.74+deb13+1-amd64

Paquetes sugeridos:
  firmware-linux-free  linux-doc-6.12  debian-kernel-handbook

Summary:
  Upgrading: 100, Installing: 2, Removing: 0, Not Upgrading: 0
  Download size: 262 MB
  Space needed: 187 MB / 14.4 GB available

Des:1 http://security.debian.org/debian-security trixie-security/main amd64 libpng16-16t64 amd64 1.6.48-1+deb13u4 [28
kB]
Des:2 http://deb.debian.org/debian trixie/main amd64 libc6 amd64 2.41-12+deb13u2 [2 847 kB]
Des:3 http://security.debian.org/debian-security trixie-security/main amd64 libfreetype6 amd64 2.13.3+dfsg-1+deb13u1
452 kB]
Des:4 http://security.debian.org/debian-security trixie-security/main amd64 libtiff6 amd64 4.7.0-3+deb13u2 [345 kB]
Des:5 http://deb.debian.org/debian trixie/main amd64 base-files amd64 13.8+deb13u4 [73.3 kB]
Des:6 http://deb.debian.org/debian trixie/main amd64 bash amd64 5.2.37-2+b8 [1 501 kB]
Des:7 http://security.debian.org/debian-security trixie-security/main amd64 imagemagick-7-common all 8:7.1.1.43+dfsg1
1+deb13u7 [73.8 kB]
Des:8 http://security.debian.org/debian-security trixie-security/main amd64 libmagickcore-7.q16-10 amd64 8:7.1.1.43+d
sg1-1+deb13u7 [1 854 kB]
3% [6 bash 595 kB/1 501 kB 40%] [8 libmagickcore-7.q16-10 710 kB/1 854 kB 38%] 119 kB/s 35min 56s
```

despues de configurar repositorios finalmente inicia la instalacion del doker

```
Paquetes sugeridos:
  cgroupfs-mount  docker-model-plugin  git-email  gitk  git-cvs  git-svn  diffutils-doc
  | cgroup-lite  git-doc  git-gui  gitweb  git-mediawiki  ed

Summary:
  Upgrading: 0, Installing: 13, Removing: 0, Not Upgrading: 0
  Download size: 106 MB
  Space needed: 418 MB / 13.9 GB available

Des:1 http://deb.debian.org/debian trixie/main amd64 pigz amd64 2.8-1 [62.7 kB]
Des:2 http://deb.debian.org/debian trixie/main amd64 liberror-perl all 0.17030-1 [26.9 kB]
Des:3 https://download.docker.com/linux/debian trixie/stable amd64 containerd.io amd64 2.2.3-1~debian.13~trixie [23.6
MB]
Des:4 http://deb.debian.org/debian trixie/main amd64 git-man all 1:2.47.3-0+deb13u1 [2 205 kB]
Des:5 http://deb.debian.org/debian trixie/main amd64 git amd64 1:2.47.3-0+deb13u1 [8 862 kB]
Des:6 http://deb.debian.org/debian trixie/main amd64 libslirp0 amd64 4.8.0-1+b1 [66.4 kB]
Des:7 http://deb.debian.org/debian trixie/main amd64 patch amd64 2.8-2 [134 kB]
Des:8 http://deb.debian.org/debian trixie/main amd64 slirp4netns amd64 1.2.1-1.1 [39.3 kB]
Des:9 https://download.docker.com/linux/debian trixie/stable amd64 docker-ce-cli amd64 5:29.4.1-1~debian.13~trixie [17
.0 MB]
39% [9 docker-ce-cli 0 B/17.0 MB 0%]
```

doker version

```
root@CN:~# docker --version
Docker version 29.4.1, build 055a478
root@CN:~#
```

aquí se hizo una prueba del docker con el hello word

```
root@CN:~#  
root@CN:~# docker run hello-world  
  
Hello from Docker!  
This message shows that your installation appears to be working correctly.  
  
To generate this message, Docker took the following steps:  
1. The Docker client contacted the Docker daemon.  
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.  
   (amd64)  
3. The Docker daemon created a new container from that image which runs the  
   executable that produces the output you are currently reading.  
4. The Docker daemon streamed that output to the Docker client, which sent it  
   to your terminal.  
  
To try something more ambitious, you can run an Ubuntu container with:  
$ docker run -it ubuntu bash  
  
Share images, automate workflows, and more with a free Docker ID:  
https://hub.docker.com/  
  
For more examples and ideas, visit:  
https://docs.docker.com/get-started/  
  
root@CN:~#
```

aquí descargo el nginx

```
root@CN:~#  
root@CN:~# docker run -d -p 8080:80 nginx  
Unable to find image 'nginx:latest' locally  
latest: Pulling from library/nginx  
801a1ad15b4e: Pull complete  
3531af2bc2a9: Pull complete  
ce776bbcdad: Pull complete  
85c66128325a: Pull complete  
4677c2a9a3d4: Pull complete  
ff048f1f2159: Pull complete  
677c63196868: Pull complete  
69989ccd189b: Download complete  
54c38c75806e: Download complete  
Digest: sha256:6e23479198b998e5e25921dff8455837c7636a67111a04a635cf1bb363d199dc  
Status: Downloaded newer image for nginx:latest  
e7632e0e7133700b1817c6f4ac6e0ab902dcd57581f32fbdf4e9ee6214741d9  
root@CN:~#
```

aquí fallo el nginx porque uso el mismo puerto que cuando descargo por lo que al usar otro puerto funciona

```
root@CN:~#  
root@CN:~# docker run -d -p 8080:80 nginx  
333a19ca7c97cdd882cb5887a0f2631e2bbe83238f3d584e9d21c1bfd600a4c  
docker: Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint heuristic_lehmann (9d48d0defc66a13ba6531cea61d577ff322eb4a19c9b0bbd6176dfa3d3beb870): Bind for 0.0.0.0:8080 failed: port is already allocated  
  
Run 'docker run --help' for more information  
root@CN:~# docker ps
```

```
root@CN:~#  
root@CN:~# docker run -d -p 8081:80 nginx  
5127f1130260c3e663635f49d8e3d27c61b9a300f91a2d578b29c064ad0889a  
root@CN:~#
```

finalmente use el docker ps

```
root@CN:~# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
5127f1130260	nginx	"/docker-entrypoint..."	About a minute ago	Up About a minute	0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
e7632e0e7133	nginx	"/docker-entrypoint..."	4 minutes ago	Up 4 minutes	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp

```
root@CN:~#
```

prueba final del funcionamiento

```
root@CN:~# curl http://10.0.2.15:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
root@CN:~# █
```